The Weil-World Supply: the non-split Weil fourfold face of $R2'$ (GST Hodge Campaign, Round 38)

GST campaign continuation session (operator order: the standing finish order with the continuation order — finish everything, no stopping after one round; `research/gpt6-native-point-primitive-span-20261017`, after Round 37 $4d6b4217$)

2026-11-05 (session clock)

Scope and the standing order

This round strikes the face (b) named by Rounds 36/37: the *non-split Weil fourfold world* — $\mathrm{NS}(A)_{\mathbb{Q}} = \mathbb{Q}\theta$, the two-dimensional Weil plane, Round 34's committed obstruction receipts. The operator's continuation order is the finish order without stopping; per the closing discipline (§40) the Clay statement is **not** claimed here, and the polarity split stands. Everything below is decided by the certificate `weil_world_supply_round38_checks.py` (182 checks, 0 failures, exit 0, deterministic double-run byte-identical, md5 of the receipts `runA.txt` (= `runB.txt`) `bdfb4d74323292a562a376ef46dad47c` (post-AUDIT-38 repair; the pre-repair receipts stood at md5 `60c3a5d374cf6b75df426a6d547974ff`), ≈13 s, exact arithmetic, zero floats). *Provenance*: the branch identifier and the Round-37 push hash are the receipts recorded in the campaign worklog (CX-17); the round artifacts live under `hodge-campaign/round38/` on the research branch.

The round's method is the campaign's letter discipline applied to the Weil world, calibrated against the repo's own committed forcing-frame receipts (the repository-root files `GSTUniversalCMArithmeticObs` and `GSTWeilForcingFrameCertificate.lean` on the research branch: the discriminating fourfold $f = [2+i]$, the spectral factors $t^2{+}14t{+}625$ / $t^2{-}30t{+}625$ / $t{-}25$, the projector $P_W = -\frac{(t-25)(39t-79)(t^2-30t+625)}{44000000}$, the correspondence coefficients $a_0, \dots, a_4$, $\mathrm{INT}\, W_R^2 = 8$, and the kernel-verified source dichotomy).

1 The Weil letter model

1.1 Setup

Let A be an abelian fourfold of Weil type over the imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$: $H^1(A, \mathbb{Q})$ is a free K-module of rank 4 (the Q-letters $e_1, \dots, e_4, f_1, \dots, f_4$ with $f_j = \sqrt{-d}\, e_j$), and the Weil-type condition on the Hodge structure ($\dim(H^{1,0} \cap V_\pm) = 2$ for the two K-eigenlines in $H^1 \otimes \mathbb{C}$) makes the degree-4 *Weil plane* W = the α^4-primary part of $[\alpha]^*$ on $\bigwedge^4_{\mathbb{Q}} H^1$ — equivalently the image of the K-line $\bigwedge^4_K H^1$ — a space of rational $(2,2)$-classes. The Hodge-type inclusion $W \subset \mathrm{Hdg}^4$ is the committed **external leaf** (inherited from the repo's forcing

frame; the certificate verifies the exact Q-arithmetic on both sides of it). The polarization $\theta = \sum_j e_j \wedge f_j$ satisfies $[\alpha]^*\theta = N(\alpha)\,\theta$.

1.2 The committed census

The nine generic worlds (K, α): the Heegner-class fields $d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$ with $\alpha = 2 + \sqrt{-d}$ at $d \in \{1, 3\}$ and $\alpha = 1 + \sqrt{-d}$ elsewhere; plus the *Eisenstein degeneracy* $(d, \alpha) = (3,\, 1 + \sqrt{-3})$ carried separately (§4).

Theorem 1.1 (S21, the spectral law)**.** *At every committed generic world, the Q-spectral structure of $[\alpha]^*$ on $\bigwedge^4_{\mathbb{Q}} H^1$ (70-dimensional) is*

$$\mathrm{minpoly}(\alpha^4)^1 \;\oplus\; \mathrm{minpoly}(\alpha^3\bar\alpha)^{16} \;\oplus\; \big(t - N(\alpha)^2\big)^{36}, \qquad 2 + 32 + 36 = 70,$$

*the eigenvalues being $\alpha^k\bar\alpha^{4-k}$ with multiplicities $\binom{4}{k}\binom{4}{4-k}$ (trace-verified against the actual 70×70 exterior matrix at every world). The three factors are pairwise coprime iff $(\alpha/\bar\alpha)^k \neq 1$ for $k \le 4$; the **Weil plane** $W :=$ the $\mathrm{minpoly}(\alpha^4)$-primary part has dimension exactly **2** at every generic world.*

Proof. The letter-pair block of $[\alpha]$ has characteristic polynomial $(x - \alpha)(x - \bar\alpha)$; the exterior power $\bigwedge^4$ of the eight Q-letters carries the products $\alpha^k\bar\alpha^{4-k}$ with the stated multiplicities, and the Q-irreducible grouping is as displayed (the conjugate pairs $k \leftrightarrow 4 - k$ merge over Q). The certificate verifies the trace identity $\sum_k \binom{4}{k}\binom{4}{4-k}\alpha^k\bar\alpha^{4-k} = \mathrm{tr}(\bigwedge^4[\alpha])$ exactly at every world, together with the three kernel dimensions 2/36/32 and the coprimality gcds. The coprimality criterion: $\mathrm{minpoly}(\alpha^4) = \mathrm{minpoly}(\alpha^3\bar\alpha)$ iff $\alpha^4 = \overline{\alpha^3\bar\alpha}$ iff $(\alpha/\bar\alpha)^3 = 1$; similarly for the other pairs. □

2 The projector supply

Theorem 2.1 (S22, the CRT projector supply)**.** *At every committed generic world there is a **unique** Q-polynomial P of degree ≤ 4 with*

$$P \equiv 1 \pmod{\mathrm{minpoly}(\alpha^4)}, \qquad P \equiv 0 \pmod{(t - N(\alpha)^2)}, \qquad P \equiv 0 \pmod{\mathrm{minpoly}(\alpha^3\bar\alpha)}.$$

*As a polynomial in the correspondence operator $[\alpha]^*$, $P([\alpha]^*)$ is an explicit Q-combination of the graphs $\Gamma_{[\alpha]^j}$ of the isogeny iterates — an **algebraic correspondence** at every committed world. At the repo's discriminator $(K, \alpha) = (\mathbb{Q}(i),\, 2 + i)$ the canonical representative equals, coefficient for coefficient, the repo's committed projector*

$$P_W(t) = -\frac{(t - 25)(39t - 79)(t^2 - 30t + 625)}{44000000} = -\tfrac{39}{44000000}t^4 + \tfrac{139}{2750000}t^3 - \tfrac{527}{400000}t^2 + \tfrac{359}{22000}t - \tfrac{79}{2816},$$

with $a_0..a_4$ the correspondence coefficients of $\Pi^{\mathrm{CH}}_W = \sum_j a_j\Gamma_{f^j}$ and the key identity $\mathrm{pwN}(t) + (t^2{+}14t{+}625)(39t^2{-}2770t{+}72375) = 44000000$.

Proof. The three factors are pairwise coprime (S21), so the CRT solution exists; two degree-≤ 4 solutions differ by a multiple of the degree-5 product, hence coincide. The certificate constructs P by modular inversion at every world and verifies the three congruences, the degree bound, and — at the discriminator — the coefficient-for-coefficient identity with the repo's P_W, the a_j list, and the key identity. □

Theorem 2.2 (S23, the divisor-sector kill and the fixation)**.** *At every committed generic world, with P the S22 projector:*

1. ***The kill**: P annihilates the 36-dimensional ambient sector and the 32-dimensional mixed sector; in particular $P(\theta^2) = 0$ — under $\mathrm{NS}(A)_{\mathbb{Q}} = \mathbb{Q}\theta$,* every product of divisor classes is killed. *The algebraic source, if it exists, must be beyond-divisor.*

2. ***The fixation**: P is the identity on W (verified on the basis and on deterministic pseudo-random combinations).*

3. ***The measurement**: the beyond-divisor content is exactly the Weil plane: $W \cap \text{ambient} = 0$ (the spectral disjointness); $\mathrm{Hdg}^4 = \mathbb{Q}\theta^2 \oplus W$ under the external leaf, $\dim W = 2$.*

4. ***The generators and the period identity** (the Gaussian world $d = 1$): $\Omega = \bigwedge_{j=1}^{4}(e_j + if_j)$ has 16 monomial components; $W = \langle \mathrm{Re}\,\Omega, \mathrm{Im}\,\Omega \rangle$; $\mathrm{INT}\, W_R^2 = 8$ (the repo's Part B receipt, reproduced by the exterior sign count).*

5. ***The fourth-power character**: the K-action on $(\mathrm{Re}\,\Omega, \mathrm{Im}\,\Omega)$ has first column $(-7, -24)$, trace -14, and determinant 625: char poly $t^2 + 14t + 625$ — $[\alpha]^*|_W$ is multiplication by $\bar{\alpha}^4 = -7 - 24i$ on the $\{\mathrm{Re}, \mathrm{Im}\}$ basis (the conjugate identification $\mathrm{Re} \leftrightarrow 1$, $\mathrm{Im} \leftrightarrow -i$; the Weil factor either way).*

6. ***The dichotomy**: $\mathrm{minpoly}(\alpha^4)$ is irreducible over Q, so the K-module $W \cong \mathbb{Q}(\alpha^4)$ is a field: any K-stable Q-subspace of W is 0 or all of W — the kernel-verified source dichotomy of the repo, reproduced in the letter model (a rank-one algebraic source is impossible; the source is zero or the full plane).*

3 The Eisenstein degeneracy

Proposition 3.1. *At $(K, \alpha) = (\mathbb{Q}(\sqrt{-3}), 1+\sqrt{-3})$ the phase $\alpha/\bar{\alpha} = e^{i\pi/3}$ is a cube root of unity: $\alpha^4 = \overline{\alpha^3\bar{\alpha}}$ exactly, the Weil and mixed factors coincide ($t^2 + 16t + 256$), and the primary part merges to dimension $34 = 2 + 32$. The degeneracy occurs at* no *generic world (the certificate's canary: $\alpha^4 \neq \overline{\alpha^3\bar{\alpha}}$ at all nine). The genericity law: the three factors are pairwise coprime iff $(\alpha/\bar{\alpha})^k \neq 1$ for $k \leq 4$ — the root-of-unity phase exclusion.*

4 The reduction and the re-drawn boundary (S24)

Theorem 4.1 (S24, the reduction to one source)**.** *Fix a committed generic world and grant the external leaf $W \subset \mathrm{Hdg}^4(A)$ (the Weil-type Hodge condition). Then the rational Hodge conjecture at A at degree 4 — $\mathrm{Hdg}^4(A) = \mathbb{Q}\theta^2 \oplus W$ algebraic — is **equivalent** to the single leaf: there exists ONE algebraic class $z \in H^4(A, \mathbb{Q})$ with $P(z) \neq 0$ (equivalently $P(z) \in W \setminus \{0\}$).*

Proof. ($\Leftarrow$) P is an algebraic correspondence (S22), so $P(z)$ is an algebraic class in W; the K-translates $[\beta]^*P(z)$ are algebraic (graphs of isogenies), and by the dichotomy (S23.6) the K-stable algebraic source they generate is the full plane W; with θ^2 algebraic, $\mathrm{Hdg}^4 = \mathbb{Q}\theta^2 \oplus W$ is algebraic. ($\Rightarrow$) trivial. The certificate verifies both halves' arithmetic: the correspondence supply, the dichotomy, and the kill (the divisor subalgebra alone can never fire the source). □

The boundary after this round: the face (b) is reduced at the committed census to the **one-source leaf** — exactly the repo's own committed open content ("the remaining nonalgebraicity claim, if pursued, must establish that $P_W \,\mathrm{cl}(Z) = 0$ for *every* actual algebraic surface Z, not just the divisor subalgebra"): either an algebraic z with $P(z) \neq 0$ is exhibited (the RHC closes at that world, and the family-transport law carries the shape), or all are killed. The residual of (a') beyond its census, $R3$ (the K-condition at general input), and $R4$ (the transcendental fold-birth) stand untouched.

The certificate

`weil_world_supply_round38_checks.py`: V1 the census (the K-operator laws at every world); V2 the spectral law (trace-verified, the $2/36/32$ kernels, the coprimality genericity); V3 the repo calibration (the three factors, the CRT projector equal to the repo's P_W coefficient for coefficient, the a_j, the key identity); V4 the projector supply at every world (the congruences, the action on W basis and random-class, the ambient and mixed kills); V5 the divisor sector ($[\alpha]^*\theta = N(\alpha)\theta$, the θ^2 eigenvalue, the divisor-sector kill, the spectral disjointness, the generators, $\mathrm{INT}\, W_R^2 = 8$, the fourth-power character, the dichotomy and its canary); V6 the Eisenstein degeneracy; V7 the hostile battery (letter-deletion, projector-sign, fake-ambient, non-word, garbage liveness, the degeneracy claim — every control fires) plus the mechanical vacuity and self-comparison scans. One defect was caught and repaired during authoring by the certificate's own scan (a dead `or True` remnant in the disjointness draft — the ledger-092 class, caught by the ledger-107 gate).

Boundary on every artifact

Q1 (the framework conjecture on the committed world model) remains proven and certified, unchanged. Q2 (the Clay statement) is **not claimed**; the closing string is **not emitted**; the polarity split stands. The falsifiable line after this round: the one-source leaf at the committed Weil census; the transport families of (a') and (b) beyond their censuses; $R3$; $R4$.